51.504 Machine Learning (2025)
Homework 2

Due 18 Nov 2025, Tuesday, 11.59pm

Question 1

We want to minimize the function $f(x, y) = y - 3x$ subject to the constraint $x^2 + y^2 = 10$.

(a) [5 points] Write down the Lagrangian for this optimization problem.

(b) [10 points] Calculate the solution of this optimization problem (The solution must satisfy the method of Lagrange multipliers since the set of possible solutions is bounded.)

(c) [5 points] Suppose now the constraint of the equation is $x^2 + y^2 \leq 10$. Write out the corresponding Lagrangian.

(d) [5 points] Write out the complementary slackness equation(s) in terms of the optimization variables x and y, and the Lagrange multiplier in part (c).

Question 2

Suppose there are four points $\{(2, 1), (1, 1), (0, 1), (-2, -1)\}$ in the (x, y)-space, where x is the input variable and y is the label variable. We will attempt to linearly separate these four points.

(a) [5 points] Suppose the initialization parameters are $\theta_1 = 2$ and $\theta_0 = -3$. Write out the equation of the initial linear separator in terms of x.

(b) [5 points] Identify the misclassified point(s) in the set of four points using this initial linear separator.

(c) [10 points] We will perform one step of the perceptron algorithm. Choose one of these misclassified points, and write out the new set of parameters after one step of gradient descent in terms of the step function α.

(d) [5 points] In this simple example, we can immediately choose the appropriate learning rate α such that the perceptron algorithm correctly classifies all the points in one step. Write down such an α.

Question 3

(a) [20 points] A Gaussian process regressor has been initialized with the kernel function

$$K(x, y) = (xy + 1)^2.$$

and the first two data points $\{(x^{(1)}, t^{(1)}), (x^{(2)}, t^{(2)})\} = \{(1, 1), (3, -1)\}$ have been sampled.

If we perform prediction, what is the mean and variance of the posterior distribution of the third point, as a function of $x = x^{(3)}$? (Hint: Calculate C^n)

(b) [5 points] Show that $K(x, y) = xy - 5$ is not a kernel function.

Question 4

For this problem, we will practice using support vector machines on the IRIS dataset. You can download the file 'IRIS.csv' off edimensions. The goal of this problem is to train an SVM model to distinguish between the three species of the IRIS data set. I did this on python but you are welcome to use either R or Matlab. There are three main goals:

(a) [5 points] Split the data into training and test set randomly, choosing a reasonable train set size.

(b) [10 points] Training and fitting the model using the training data.

(c) [10 points] Predicting the test set using the SVM regions (calculate a confusion matrix and classification report, as in how many were correctly classified).

The IRIS data set consists of 100 data points, 4 input variables and a output species variable with three states. You can choose to perform SVM on a subset of these data as long as the three steps above are clearly stated. That means, you can choose to do it for either only 2 input variables so it is easy to visualize, or only 2 species instead of three. SKlearn on python has the ability to classify all three classes, so go ahead with whichever you feel comfortable with, as long as the above three main goals are satisfied.